

P2: Sequence and Data

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Overview

Now we can start doing some programming of our own. In this task the focus is on:

1. Creating small programs that involve a sequence of instructions, and introduce much of the main programming terminology.
2. Extending your programs with variables to manage data, and functions to calculate things.

Tip: Related Credit Glossary Tasks

If you are aiming for Credit or higher, make sure to be updating your glossary as you go.

Do the following:

Part 1: Sequence

- Everything you need is in the [Sequence](#) chapter of our [Programmer's Field Guide](#). The [Intro Tours](#) walk through the concepts, and should give you a good start.

Tip: What is studying?

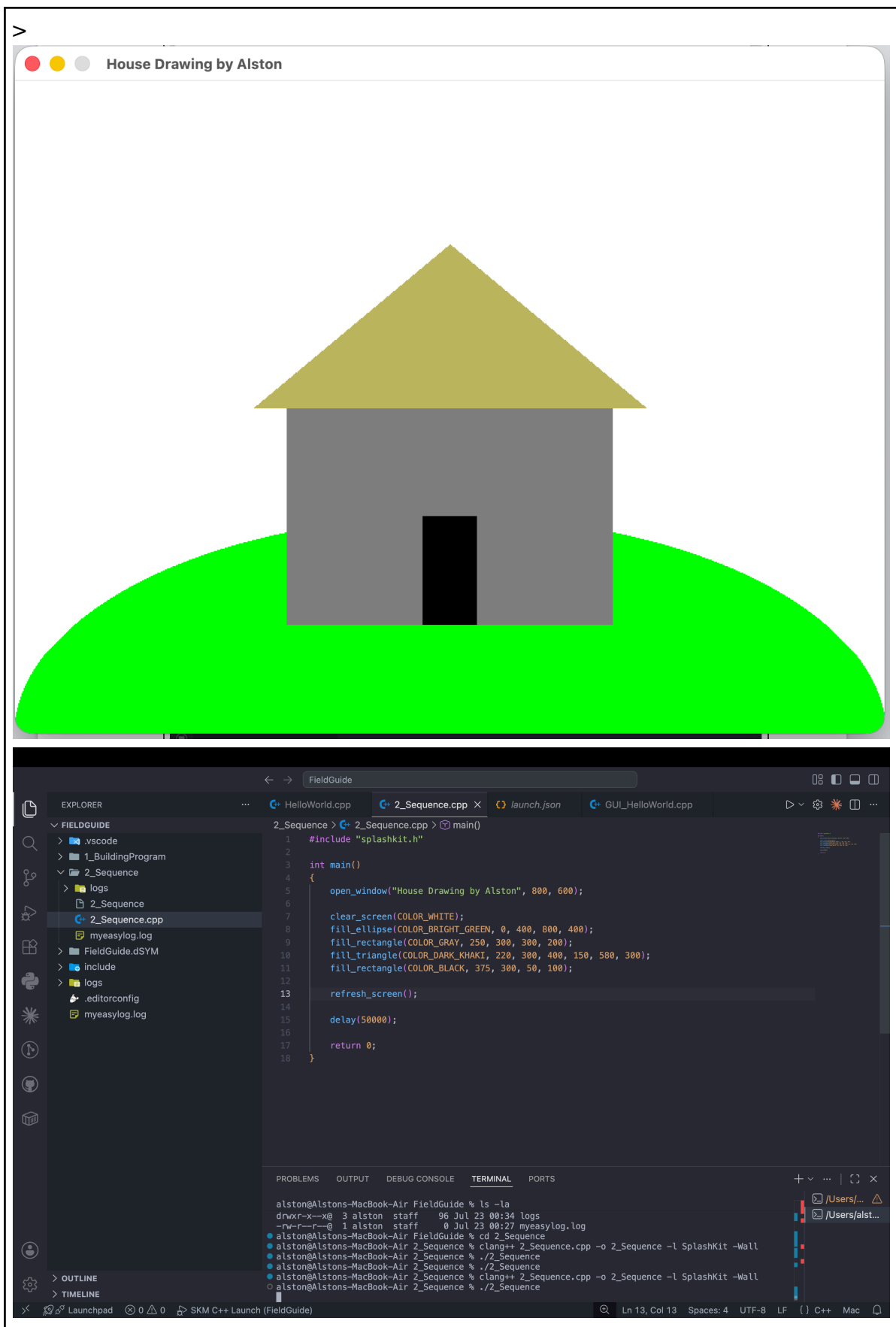
Please make sure to **study** the material in the Programmer's Guide, not just read/watch it.

We created the Programmer's Field Guide specifically for this unit to help you develop your programming capabilities. But **reading is not learning!** You need to engage with the material **actively**. This means taking notes (in whatever form works for you, but make sure to check out the C1 task), trying out the code snippets, and discussing the concepts with others. Beware of the trap of *feeling* productive without actually *being* productive.

Tip: Exercising your agency in these tasks

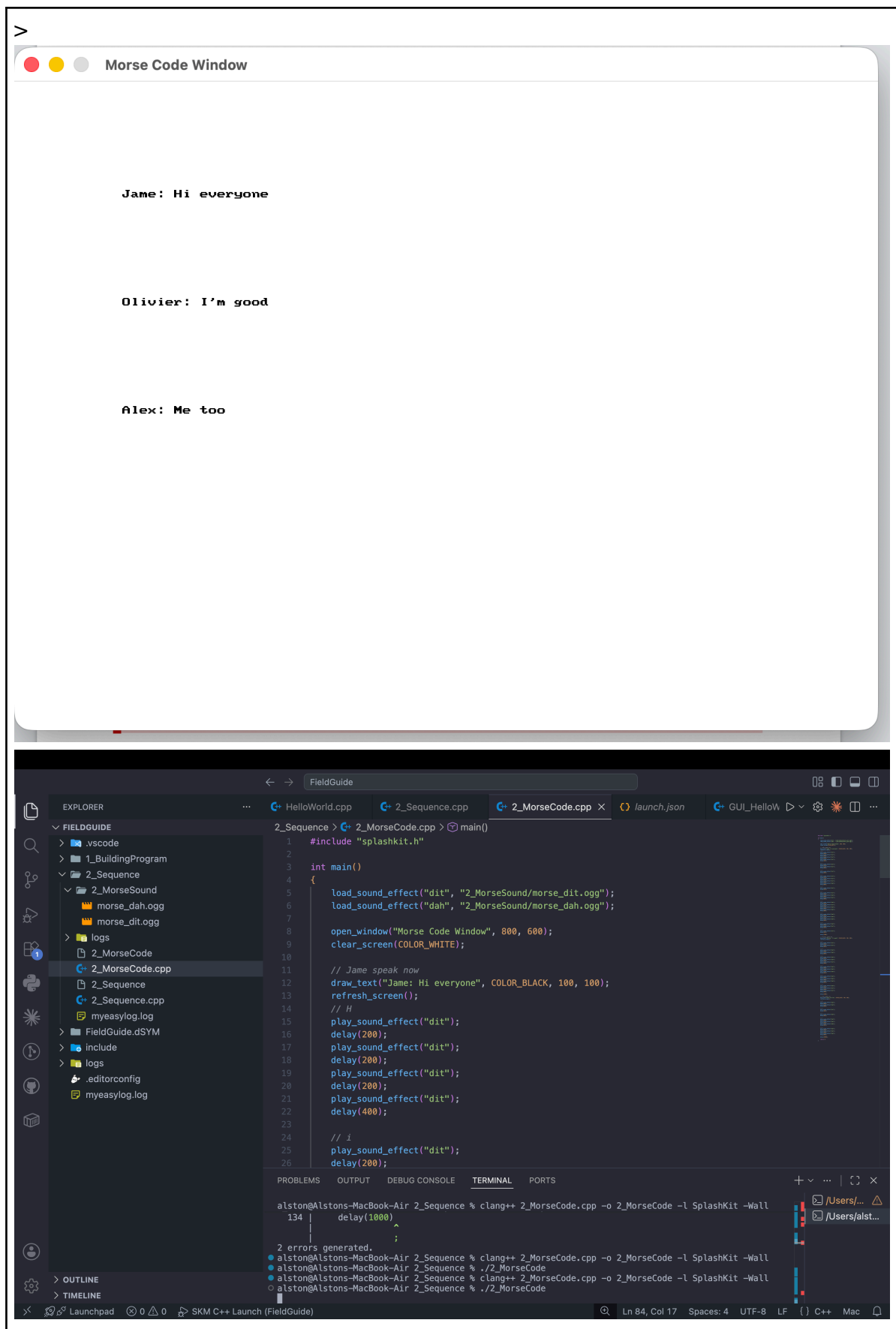
When we created the Field Guide, we added guided tours to help walk you through applying the ideas covered. Each of the worksheets will recommend you work through a number of these tours. If you want to do something else, you can do that. We just ask that you discuss this with the teaching team first. That way we can help ensure that what you do will help you develop the required capabilities.

- Add a screenshot showing you were able to get a program to draw something using a sequence of calls. You can use the [house drawing tour](#)! It will be very similar to the guide, that's OK.



- Complete one of the [Test Your Capabilities](#) tasks from the chapter to see that you can now put these tools together to achieve something yourself.

>First, I want to create a list include all morse code, then when I want to print some word, I can output the sound of morse code easily by using the list. But it to difficult to do it. So I change the mind. I use `open_window()` to have a space to print the word that the people want to say, my plan is print the word and then play the morse code sound dit & dah corresponding, by this , I need to use `clear_screen()` and `refresh_screen()` carefully. I use `draw_text()` to print the word on the window and I give the wrong parameter initially, but the terminal show me what was the parameter is `draw_text()` needed, so I can do it well. The another problem is about the delay first I just put `delay(100)` and `delay(300)`, I forget "dit" and "dah" have playing time 100ms and 300ms , so the I fix the problem.



Caution: Avoid undermining your learning by offloading it to others

AI is great at coding! But remember that you are here because *you* want to learn how to program. So avoid getting AI to write the code for your submissions.

No need to use AI to do the reflections for you either - make these personal. We do not want these to be formal, polished pieces of work. A rough, personal account will be preferred to one that is highly refined! Just show us what you did, and don't waste your time trying to make this look impressive.

Part 2: Data

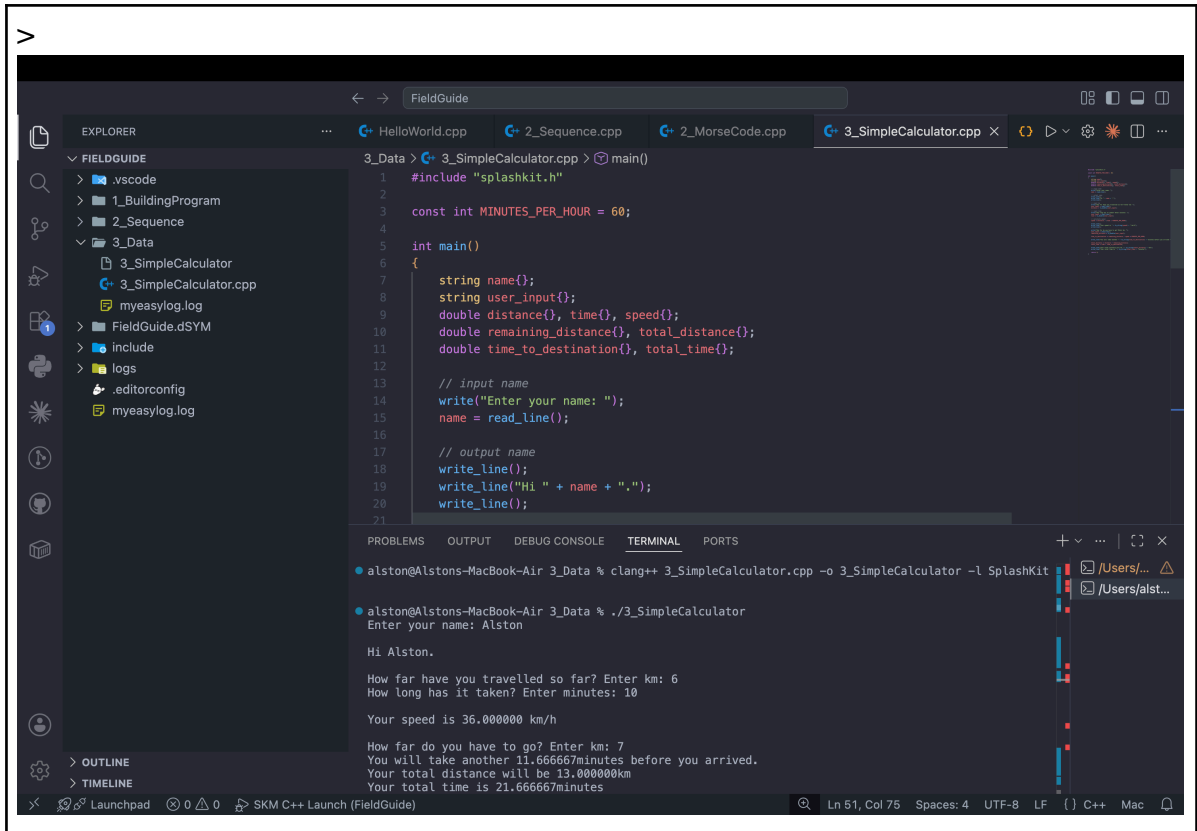
- Everything you need is in the [Data](#) chapter of our [Programmer's Field Guide](#). The [Intro Tours](#) walk through the concepts, building on what was covered in the sequence chapter.

Tip: Coding style and language use

Make sure to follow our [style guide](#). This will be consistent with the code you see in our field guide, so try to match our examples. OnTrack will check this when you submit.

Also ensure that you are only using concepts covered so far. We want you to focus just on these concepts and how they work together.

- Add a screenshot showing you were able to build a program that performs a sequence of calculations. You can use the [simple calculator tour](#)! Show VS Code with the terminal output, and your code.

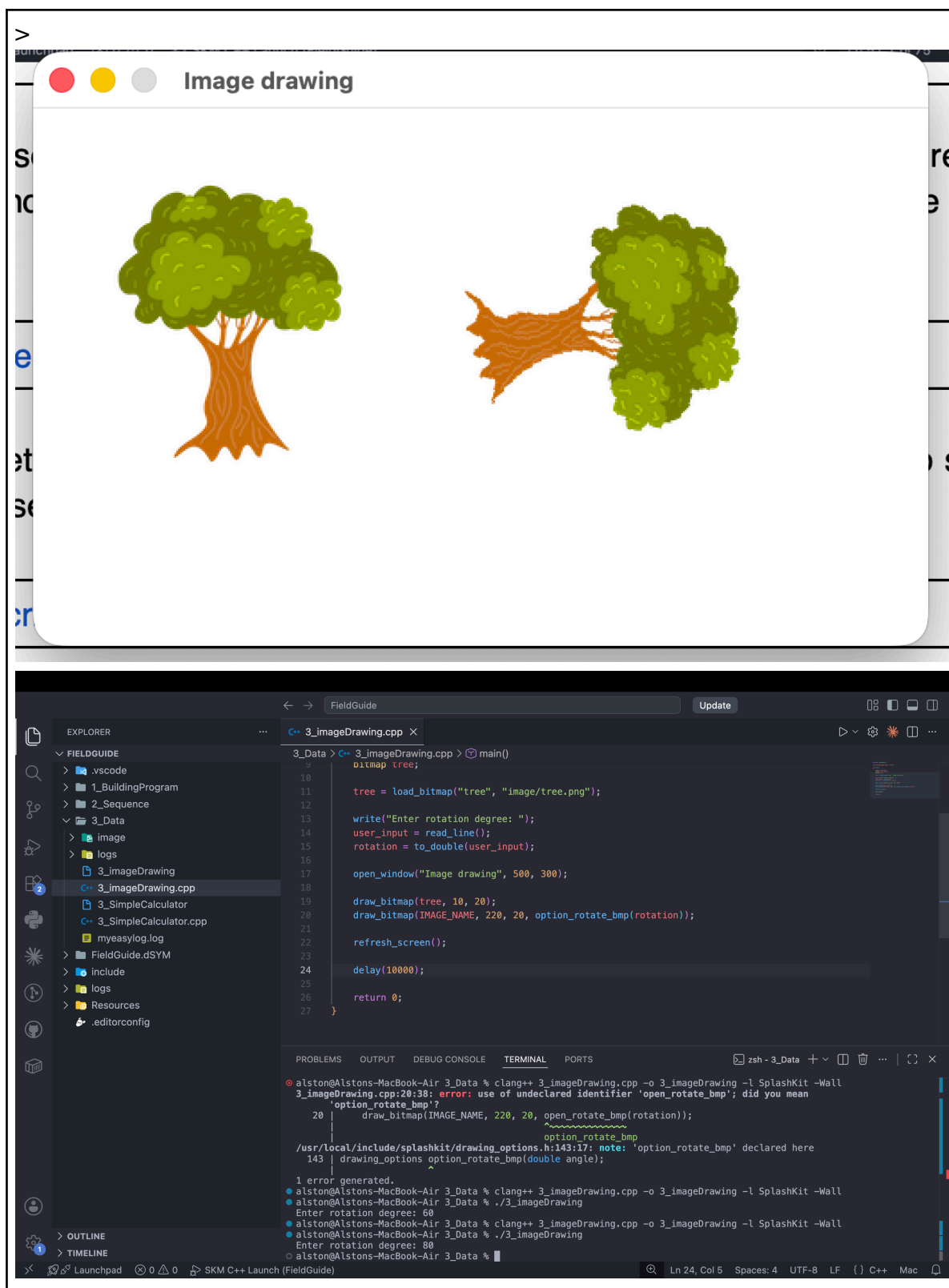


The screenshot shows a Visual Studio Code editor window with the file explorer on the left and the code editor in the center. The file explorer shows a project named 'FieldGuide' with a subdirectory '3_Data' containing '3_SimpleCalculator.cpp'. The code editor displays the contents of '3_SimpleCalculator.cpp', which includes a header file 'splashkit.h' and a main function. The main function prompts the user for their name, distance, and time, and then calculates and displays the speed, remaining distance, and total time. The terminal at the bottom shows the output of the program, which matches the expected behavior described in the text.

```
>  
FieldGuide  
3_Data > 3_SimpleCalculator.cpp > main()  
1 #include "splashkit.h"  
2  
3 const int MINUTES_PER_HOUR = 60;  
4  
5 int main()  
6 {  
7     string name{};  
8     string user_input{};  
9     double distance{}, time{}, speed{};  
10    double remaining_distance(), total_distance{};  
11    double time_to_destination(), total_time{};  
12  
13    // input name  
14    write("Enter your name: ");  
15    name = read_line();  
16  
17    // output name  
18    write_line();  
19    write_line("Hi " + name + ".");  
20    write_line();  
21
```

alston@Alstons-MacBook-Air 3_Data % clang++ 3_SimpleCalculator.cpp -o 3_SimpleCalculator -l SplashKit
alston@Alstons-MacBook-Air 3_Data % ./3_SimpleCalculator
Enter your name: Alston
Hi Alston.
How far have you travelled so far? Enter km: 6
How long has it taken? Enter minutes: 10
Your speed is 36.000000 km/h
How far do you have to go? Enter km: 7
You will take another 11.666667minutes before you arrived.
Your total distance will be 13.000000km
Your total time is 21.666667minutes

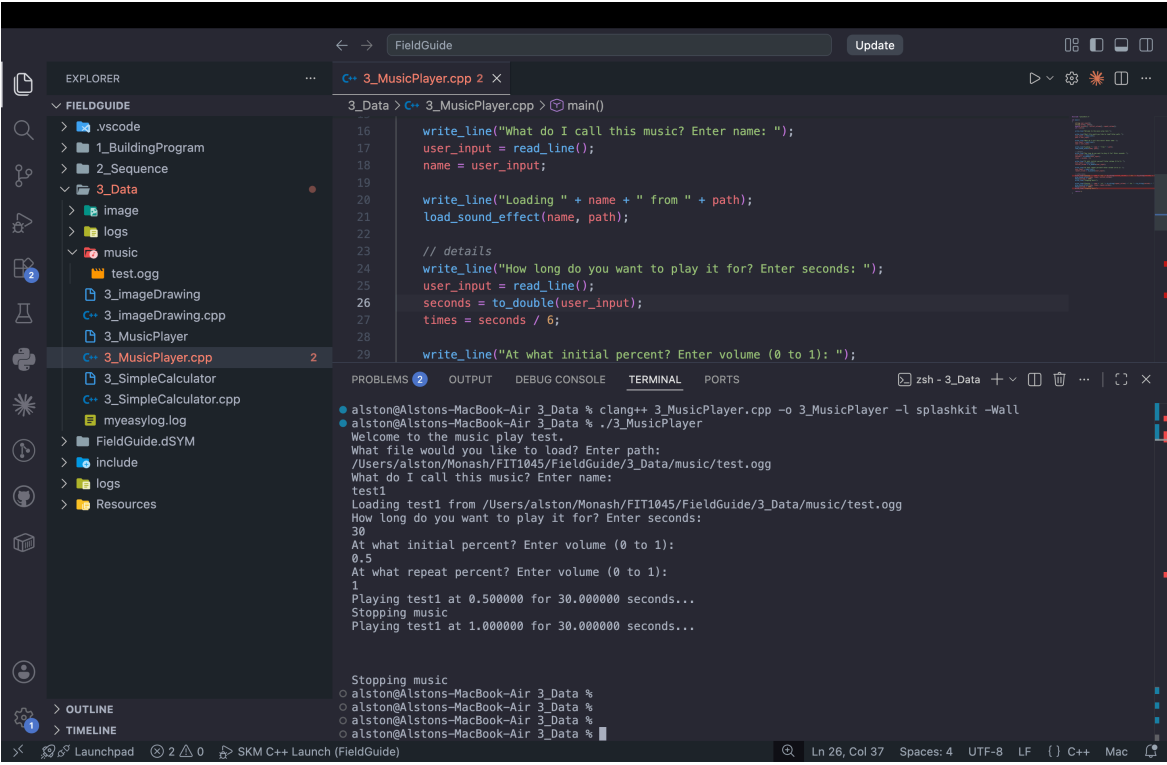
- Add a screenshot showing you were able to build a program that reads values from the user and uses these to draw something to the screen. You can use the [dynamic drawing tour](#)!



- Complete one of the [Test Your Capabilities](#) tasks from the chapter to see that you can now put these tools together to achieve something yourself.

> I choose "Music Player". First, I use `write_line()` write out everything need to output , and then use `user_input = read_line()` read everything need to input. Initially I let `user_input_string` and `user_input_double`, but later I remember about I can use `to_double(user_input)` then I change all to more proper way. One of the problem I face is about the "How long do you want to play the music? ", I let the variable call "seconds", and one be "times", first I like to find a function what can detect how long is the music what user enter, but I failed, so I use `times = seconds / 6` manually, which times is defined by `int times{}`, make sure is integer and correct in `play_sound_effect()`. And the last issue is I need to use `to_string()` to print the sentences.

>



```

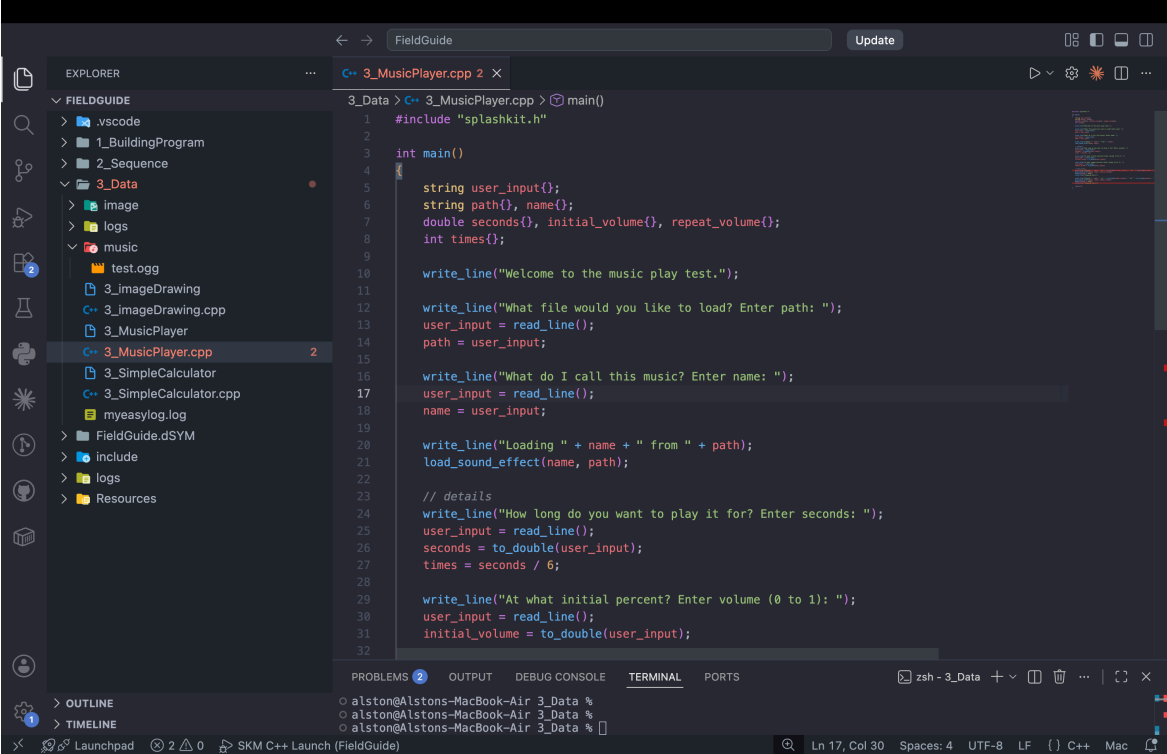
3_Data > C++ 3_MusicPlayer.cpp > main()
16  write_line("What do I call this music? Enter name: ");
17  user_input = read_line();
18  name = user_input;
19
20  write_line("Loading " + name + " from " + path);
21  load_sound_effect(name, path);
22
23  // details
24  write_line("How long do you want to play it for? Enter seconds: ");
25  user_input = read_line();
26  seconds = to_double(user_input);
27  times = seconds / 6;
28
29  write_line("At what initial percent? Enter volume (0 to 1): ");

```

```

alston@Alstons-MacBook-Air 3_Data % clang++ 3_MusicPlayer.cpp -o 3_MusicPlayer -l splashkit -Wall
alston@Alstons-MacBook-Air 3_Data % ./3_MusicPlayer
Welcome to the music play test.
What file would you like to load? Enter path:
/Users/alston/Monash/FIT1045/FieldGuide/3_Data/music/test.ogg
What do I call this music? Enter name:
test1
Loading test1 from /Users/alston/Monash/FIT1045/FieldGuide/3_Data/music/test.ogg
How long do you want to play it for? Enter seconds:
30
At what initial percent? Enter volume (0 to 1):
0.5
At what repeat percent? Enter volume (0 to 1):
1
Playing test1 at 0.500000 for 30.000000 seconds...
Stopping music
Playing test1 at 1.000000 for 30.000000 seconds...
Stopping music
alston@Alstons-MacBook-Air 3_Data %
alston@Alstons-MacBook-Air 3_Data %
alston@Alstons-MacBook-Air 3_Data %
alston@Alstons-MacBook-Air 3_Data %

```



```

1  #include "splashkit.h"
2
3  int main()
4
5      string user_input();
6      string path(), name();
7      double seconds(), initial_volume(), repeat_volume();
8      int times();
9
10     write_line("Welcome to the music play test.");
11
12     write_line("What file would you like to load? Enter path: ");
13     user_input = read_line();
14     path = user_input;
15
16     write_line("What do I call this music? Enter name: ");
17     user_input = read_line();
18     name = user_input;
19
20     write_line("Loading " + name + " from " + path);
21     load_sound_effect(name, path);
22
23     // details
24     write_line("How long do you want to play it for? Enter seconds: ");
25     user_input = read_line();
26     seconds = to_double(user_input);
27     times = seconds / 6;
28
29     write_line("At what initial percent? Enter volume (0 to 1): ");
30     user_input = read_line();
31     initial_volume = to_double(user_input);
32

```

```

alston@Alstons-MacBook-Air 3_Data %
alston@Alstons-MacBook-Air 3_Data %
alston@Alstons-MacBook-Air 3_Data %

```

```

33  write_line("At what repeat percent? Enter volume (0 to 1): ");
34  user_input = read_line();
35  repeat_volume = to_double(user_input);
36
37  // play music
38  write_line("Playing " + name + " at " + to_string(initial_volume) + " for " + to_string(seconds) + " seconds");
39  play_sound_effect(name, times, initial_volume);
40  delay(seconds * 1000);
41  write_line("Stopping music");
42

```

Reflect:

- What was the most interesting thing you learnt through this task, and why?

> P2 is more complex task compare to P1!! I learned more deeply about the sequence and data, especially data types, it's very important, if not careful about that, it can cause many bug.

- To enable you to better reflect on your performance over the long term in this unit, fill in this [google form](#).
- Looking back, how long did it take to complete this task? What did you spend the most time on?

> The guide tour part just need little time to complete. But the morse code and music player both take me 3-4 hours to thinking, building program, debug and run.

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I did not use any AI for this task.

Submit:

- Export this worksheet as a PDF.
- Upload your submission to OnTrack:
 - Mark the task as **Ready for Feedback** in OnTrack and upload the required files.
- Engage with feedback and discussion to get this signed off.
 - To get the task **Complete**, you need to engage with the feedback you receive. Your tutor will review your submission and may ask you to clarify aspects of the task or include aspects you have missed.
 - Remember, a **Resubmit** status is not a bad thing – your tutor is just trying to help improve your understanding. When you do resubmit, make sure to include a comment that describes what you changed. This needs to demonstrate how you have addressed all of your tutor's feedback.

Tip: Initiate feedback

When you submit your work, you can add a comment to your tutor. It is a great idea to use this to ask for specific feedback on your work. This can help make sure you receive the most effective feedback to support your learning.